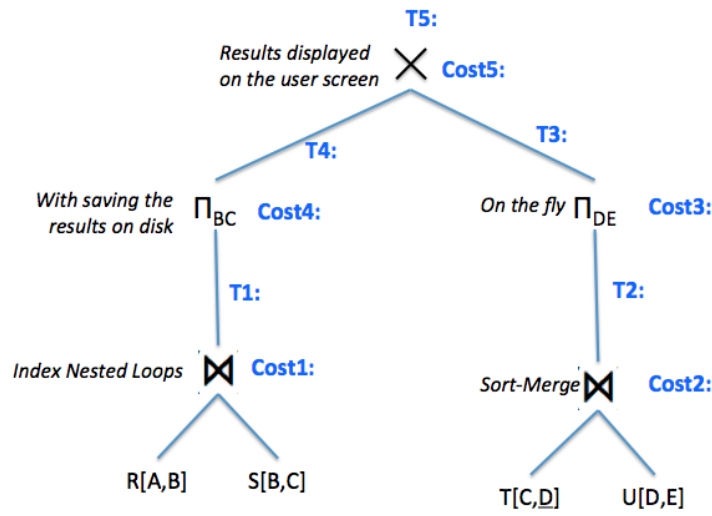


Assume that there is a B+tree on R.B, the R.B has a selectivity factor f_b , the S.B has 50% of its tuples with null value, T.D is a key for T, and U.D has no null values. Assuming the follow query plan, compute the cost of every operator in it and the number of tuples that each operator generates



Answer:

$$\sim Cost1 = P_S + |S| * \frac{50}{100} * (3 + f_B * |R|), \text{ (where } |S| = P_S * \lfloor \frac{P}{t_s} \rfloor \text{)}$$

$$T1 = |S| * \frac{50}{100} * f_B * |R|, \text{ (where } |S| = P_S * \lfloor \frac{P}{t_s} \rfloor \text{)}$$

$$Cost2 = 2P_t \log P_T + 2P_u \log P_u + P_T + P_U,$$

$$T2 = C_u,$$

$$Cost3 = 0,$$

$$T3 = T2 = C_u,$$

$$Cost4 = \frac{T1}{\lfloor \frac{P}{t_s} \rfloor}, \text{ (Cost to save the results on the disk to be used by the } \times \text{)}$$

$$T4 = T1,$$

$$Cost5 = Cost4 * \frac{T3 * t_u}{B-1}, \text{ (where } T3 * t_u \text{ is the space of the } T3 \text{ tuples, and } B \text{ is the number of buffers available, and we put } Cost4 \text{ because we need to read the } T4 \text{ tuples from the disk where we saved them)}$$

$$T5 = T3 * T4$$